

Predicting High Movie Ratings

Kueski Data Science Challenge

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Point-in-Time Classification with MovieLens 20M

github.com/4ngelGtz/movie_rating_predictor

MovieLens 20M

20M ratings

138K users

26.7K rated movies

Jan 1995 – Mar 2015

Movie catalog information and Genome content relevance data enrich rating histories.

Predict the rating outcome

For an observed rating event (u, m, t) :

$$y = \mathbf{1}\{r \geq 4\}, \quad \hat{p}(u, m, t) \approx P(y = 1 \mid \mathcal{I}_t)$$

$\mathcal{I}_t$: information available before t .

User u + Movie m + Prior information



$$P(\text{rating} \geq 4)$$

Objective

Estimate the probability that a user assigns a rating ≥ 4 given the information available before the event.

Point-in-time evaluation without temporal leakage

1995–2011

2012–2013

2014–2015

TRAIN

VALIDATION

TEST

Fit the model

Select the best iteration

Evaluate the frozen model

Strict-prior history

$$H(t) = \{e : t_e < t\}$$

History before t $\longrightarrow$ Build features $\longrightarrow$ Predict events at t



Observe outcomes



Update history with the complete batch at t

Key idea

Every prediction recreates what could have been known at that moment.

Current, simultaneous, and future targets are excluded.

Static catalog and Genome metadata are treated as available at scoring time; their historical availability is not reconstructed.

What information does the model use?

USER

How does this user normally rate?

- Rating count
- Mean rating
- Rating dispersion
- Recency

Captures the user's rating baseline, activity, and consistency.

MOVIE

How is this movie being received?

- Historical count, mean, dispersion
- Recent activity and sentiment (30/60 days)
- Genre breadth
- Release year / age
- Genome descriptors

Captures long-term and recent audience reception, plus movie context.

USER $\times$ MOVIE

Does this movie match the user's historical taste?

- User-genre affinity
- Genome semantic similarity
- Positive / negative profile similarity
- Preference margin
- Nearest liked content

Captures personalized fit between the movie and the user's prior preferences.

Shared context: global rating count and global mean.

How does the model combine these signals?

USER BASELINE

What does the user bring to the prediction?

Historical rating level
Activity and recency
Rating consistency

The same movie can receive a different score for a generous versus a strict rater.

MOVIE EVIDENCE

What does the movie bring to the prediction?

Historical reception
Recent 30/60-day sentiment
Catalog context

Long-term and recent audience response provide context around the target movie.

PERSONALIZED FIT

What changes for this specific user–movie pair?

Genre affinity
Genome semantic similarity
Preference margin
Similarity to previously liked content

The model measures whether the movie resembles what this user has historically rated highly.

User signals + Movie signals + Personalized fit

XGBoost

$P(\text{rating} \geq 4)$

XGBoost learns nonlinear interactions between user behavior, movie reception, and personalized taste.

How well does the model generalize over time?

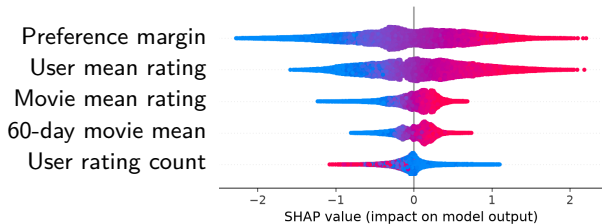
Split	Period	Events	Positive rate	AP	ROC-AUC
Train	1995–2011	17,822,773	49.8%	0.81	0.82
Validation	2012–2013	1,330,716	52.0%	0.82	0.82
Test	2014–2015	846,774	50.2%	0.81	0.82

No clear signs of overfitting; the model generalizes across the evaluated periods.

Positive rate = share of ratings ≥ 4 . AP = Average Precision.
Observed temporal stability does not guarantee future stability.

What did the model learn?

Top five SHAP signals; 50,000 test events



Color = feature value; horizontal position = SHAP contribution to model log-odds.

The model learned that personalized taste, user rating behavior, and movie reputation are all important drivers of a high rating.

PERSONAL FIT MATTERS

The strongest signal measures whether the movie is closer to the user's high-rated or lower-rated history.

USERS HAVE DIFFERENT RATING BASELINES

A four-star rating has different meaning for a generous versus a strict rater.

AUDIENCE RECEPTION STILL MATTERS

Long-term and recent movie sentiment complement personalized taste.

HISTORY DEPTH CHANGES CONTEXT

The amount of prior user history also influences the score.

How is personalized taste measured?

$$\text{Preference Margin} = \cos(g_m, p_u^+) - \cos(g_m, p_u^-)$$

Two taste profiles, built from this user's history

Movies rated ≥ 4

Positive taste
profile p_u^+

Movies rated < 4

Lower-rated
profile p_u^-

Compare cosine
similarity

Target movie g_m

g_m : target movie's Genome vector. Each profile aggregates Genome vectors of movies this user rated before t .

Margin > 0 : closer to historically high-rated content.

Margin < 0 : closer to the user's lower-rated history.

Other strong signals

User mean rating

Typical rating level of the user.

Movie mean rating

Long-term audience reception.

60-day movie mean

Recent audience sentiment.

User rating count

Amount of historical behavior observed.

- **Strong preference prediction under a point-in-time setup.**

Estimates whether an observed rating will be ≥ 4 using information before t .

- **Complementary signals explain user preference.**

Personalized taste, user rating behavior, and movie reputation all add signal.

- **Preference is not purchase propensity.**

A user may enjoy a movie and still choose not to buy or rent it. Price, promotions, availability, alternatives, and context also matter.

- **Preference is an input to the business decision.**

The score estimates user–movie affinity and should feed a downstream decision model, rather than serve as the final business objective.

Use predicted affinity inside the business decision

Proposed decision layer: not yet implemented

Preference model $\longrightarrow$ Purchase propensity $\longrightarrow$ Expected profit

$$\hat{q}(u, m, t) = P(\text{rating} \geq 4 \mid u, m, I_t)$$

$\hat{q}$ = predicted user–movie affinity $\longrightarrow$ input to the purchase model

$$P(\text{purchase} \mid u, m, t, \text{price}, \hat{q}(u, m, t))$$

Preference answers “Will they like it?”; the decision model answers
“Will they buy or rent it at this price?”

$$\begin{aligned} \text{Expected Profit}(u, m, \text{price}) &= P(\text{purchase} \mid u, m, t, \text{price}, \hat{q}(u, m, t)) \\ &\quad \times (\text{price} - \text{cost}) \end{aligned}$$

$$\text{price}^* = \underset{\text{price}}{\operatorname{argmax}} \text{Expected Profit}(u, m, \text{price})$$

The model tells us what users are likely to value; the business decision determines what to offer, to whom, and at what price.

DATA

- Validate historical availability of static catalog / Genome metadata.
- Collect purchase or rental outcomes.
- Include price, promotions, offer context, and availability.

MODEL

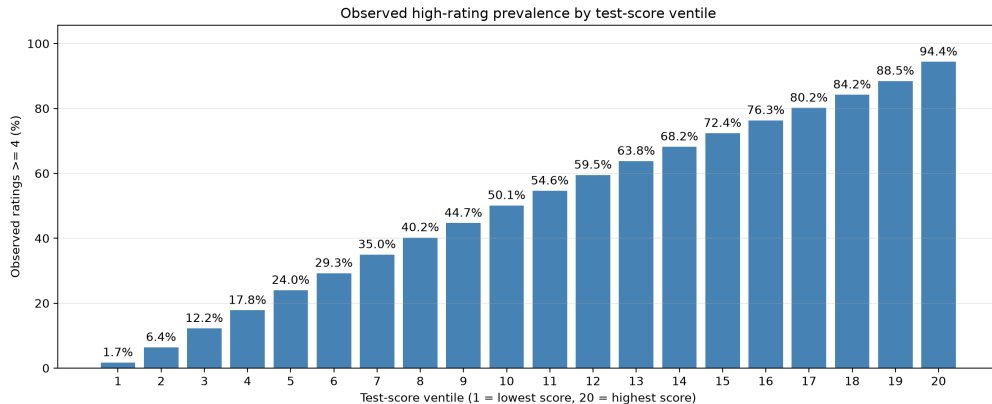
- Validate the preference model on fresh temporal data.
- Build purchase propensity using predicted affinity, price, and context.
- Assess calibration if probabilities will drive decisions.

DECISION

- Define the optimization target: profit, revenue, retention, or engagement.
- Estimate response to price and offers.
- Validate the final policy online before claiming business lift.

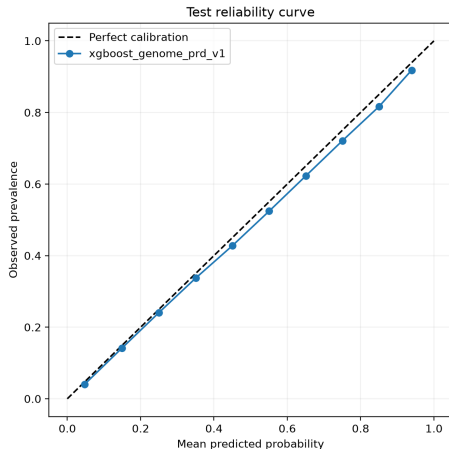
The next step is to optimize the decision, not just the prediction.

Backup: score separation



Observed high-rating prevalence rises from 1.7% to 94.4% across 20 test-score ventiles.

Backup: calibration



Expected Calibration Error
 $\approx 2.1\%$

About 2.1 percentage points.

Predictions modestly exceed observed high-rating rates in all ten equal-width test bins.